When Shared Memory Hurts: Stale-Coordination Failure in Multi-Agent Particle Collection under a Rotating Latent Field

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Submission Id: «OpenReview submission id»

Abstract

Sharing observations across a team of agents improves coordination under a stationary transport field — but what happens when the field drifts? A team that pools a long observation history can converge confidently on a stale consensus, driving all collectors toward the wrong direction while independent agents, sampling noisier but more diverse estimates, collectively cover more ground. This paper asks where exactly that reversal occurs and what governs it.

Working in a stochastic particle-collection task with a confirmed stationary coordination benefit ($\bar{\Delta} = +1.19$ particles, n=32 matched seeds), we introduce a linearly rotating field and measure the last memory window $L_{\max}$ that maintains a statistically significant benefit, across an order-of-magnitude range of rotation speeds and eight window lengths.

The central result is a *stale-coordination failure mechanism*: the shared arm reduces angular noise but leaves angular bias unchanged, so beyond a critical window length it drives all collectors confidently toward a systematically wrong direction, eliminating the coverage diversity that makes coordination beneficial. The significance-based crossover is consistent with $L_{\max} \approx c\, T_{\text{corr}}$ (where $T_{\text{corr}} = 1/(\omega \cdot dt)$ is the field's one-radian decorrelation timescale) with c ranging from 0.15 (Holm-corrected) to 0.30 (uncorrected) at two interior rotation speeds; a third tested speed did not reproduce a sharp crossover, and c should be read as a preliminary point estimate. The exponential-decay controller does not show the same pattern.

CCS Concepts

• **Computing methodologies → Multi-agent planning**; *Cooperative and collaborative learning*.

Keywords

multi-agent coordination; non-stationary environments; memory window; communication benefit; stochastic particle collection; rotating latent field

ACM Reference Format:
Anonymous Author(s). 2027. When Shared Memory Hurts: Stale-Coordination Failure in Multi-Agent Particle Collection under a Rotating Latent Field. In *Proc. of the 26th International Conference on Autonomous Agents and Multiagent Systems (AAMAS 2027), Hanoi, Vietnam, 3–7 May 2027*, IFAAMAS, 8 pages.

Proc. of the 26th International Conference on Autonomous Agents and Multiagent Systems (AAMAS 2027), M. Baldoni, F. Fang, W. Yeoh, N. Yorke-Smith (eds.), 3–7 May 2027, Hanoi, Vietnam.

1 Introduction

Communication protocols for multi-agent systems are typically designed and evaluated against a fixed environment. In deployment, environments drift. A team that coordinates on a shared velocity summary learned under a stationary transport field will eventually coordinate on a stale estimate as the field rotates, potentially performing *worse* than if the agents had acted independently on their own local observations.

This paper asks a precise question: how long must an observation memory window be to maintain a positive coordination benefit when the latent transport field rotates at angular rate ω? We study this in the stochastic particle-collection task of SPS-C03 [1], where four mobile collectors must exploit a weak latent drift α=0.06 to collect more of N=256 diffusing particles than a capacity-matched independent team with no shared information. The confirmed stationary baseline is $\bar{\Delta}$=+1.19 particles (SD 2.44, p<0.001, 32 matched seeds).

We extend this baseline along the temporal axis by introducing a linearly rotating field $\theta(t) = \theta(0) + \omega \cdot t \cdot dt$ and asking two implementation-level questions: (i) what memory window L should agents use, and (ii) how does the best achievable L scale with the rotation rate ω?

Main result. At two interior rotation speeds ($T_{\text{corr}} \in \{33, 10\}$), the pre-registered significance-based threshold satisfies

$$L_{\max}(\omega)/T_{\text{corr}} \approx 0.30, \quad T_{\text{corr}} = \frac{1}{\omega \cdot dt},$$

where T_{corr} is the one-radian decorrelation timescale. At these two speeds the window spans ≈ 0.30 rad of total field rotation at the threshold. The pattern is preliminary (see Section 5.2). The constant $c \approx 0.30$ is preliminary; the borderline slow anchor (p=0.049) motivates further validation. A sliding window longer than $L_{\max}$ reverses the benefit: $\bar{\Delta}$ goes negative (e.g. from +0.94 at L=10 to −0.38 at L=20 for ω=1.5 rad/step).

Why memory eventually hurts. The mechanism is not a degradation of each agent's individual directional accuracy. Both arms accumulate the same angular lag ($\omega \cdot L \cdot dt/2$ radians at window midpoint); the shared arm merely reduces angular *noise* while leaving the bias unchanged. Once the window spans enough rotation, the shared arm drives all M collectors confidently toward the same stale direction, eliminating the coverage diversity that makes independent estimates collectively useful. This is a policy nonlinearity: the capacity-matching controller maps confident-but-wrong directions into globally poor coverage.

Contributions.

(1) We demonstrate a stale-coordination failure mechanism: confidence in a shared but stale direction eliminates the coverage diversity of independent agents, reversing the coordination

benefit. This is well-supported in the slow rotation condition and observed, with less sharpness, in a second interior speed.
(2) We report a preliminary point estimate: the significance-based threshold satisfies $L_{\max}/T_{\mathrm{corr}} \approx 0.30$ (uncorrected) to 0.15 (Holm-corrected) at two interior speeds, with a third tested speed not reproducing a sharp crossover. This is an exploratory pattern, not a settled scaling law.
(3) We show that the empirical crossover occurs at lower $L_{\max}$ than the bias/noise-parity prediction ($L^* \propto (\omega\, dt)^{-2/3}$), a different functional form, attributable to policy nonlinearity.
(4) We show that exponential decay does not follow the same $L_{\max} \propto T_{\mathrm{corr}}$ pattern at the two interior speeds, making the sliding window a more tractable choice in the studied regime.
(5) We provide a pre-registered, reproducible experiment with event-keyed matched counterfactuals and an explicit reproduction gate anchored to SPS-C03.

2 Related Work

Non-stationary multi-agent environments. Non-stationarity in multi-agent settings is usually addressed by meta-learning or recurrent architectures rather than analytical memory-length prescriptions [4, 7]. These approaches adapt L implicitly through learned gates but offer no closed-form guidance for practitioners. Patiño et al. [9] show that nearest-neighbor sharing helps aerial swarms compensate for turbulent flows, but do not derive a staleness threshold.

Windowed and discounted estimation. Sliding-window UCB (SW-UCB) [5] and discounted UCB [8] apply the same bias/variance logic to non-stationary bandit problems: a window of length $L \approx \sqrt{n/\Delta}$ is near-optimal under a known breakpoint rate Δ. Our sliding-window controller is the cooperative analogue of SW-UCB applied to field-direction estimation across agents. We find a pattern consistent with $1/\omega$ scaling at two interior rotation speeds ($L_{\max}/T_{\mathrm{corr}} \approx 0.30$), steeper than the bias/noise-parity prediction ($L^* \propto \omega^{-2/3}$); this pattern is preliminary and requires further validation.

Communication under information constraints. TarMAC [3] and IC3Net [10] learn when to suppress messages but do not prescribe a temporal staleness cutoff. The information-bottleneck approach of Wang et al. [12] optimises message bandwidth but not memory depth. Our contribution is orthogonal: we hold the message fixed (a count-weighted team mean velocity) and vary only the observation history depth.

Particle collection. Wang et al. [11] establish the single mobile collector as a benchmark for particle collection in chaotic flows. Our multi-agent extension isolates whether shared summaries lower the minimum signal strength needed to exploit a latent field, and the current paper asks how robustly that benefit persists under field rotation. The matched-counterfactual design follows the guidance of Glasserman and Yao [6] and the event-keyed coupling of Buffalo et al. [2].

3 Model and Theory

3.1 Particle-collection task

The arena is the unit square $[0, 1]^2$ with reflecting boundaries. N=256 passive particles diffuse with step size σ=0.06 per dt=0.02 second and are advected by a latent field with strength α=0.06. Four collectors (sensing radius r=0.16, max speed $v_{\max}$=0.12) run for H=67 steps; a particle is permanently captured when it enters a collector's disc. The estimand is

$$\Delta_s = Y_{\mathrm{shared},s} - Y_{\mathrm{indep},s},$$

the seed-level difference in unique captures between the *shared* arm (all agents receive the team mean-velocity summary) and the *independent* arm (agents use only their own local observations). Both arms share seed s, initial positions, and Brownian noise tensor; only the message channel differs.

3.2 Rotating-field model

The latent field direction evolves deterministically:

$$\theta(t) = \theta(0) + \omega \cdot t \cdot dt, \quad \theta(0) \sim \mathrm{Uniform}[0, 2\pi).$$

At ω=0 the task reduces to SPS-C03 exactly. The field rotates deterministically, so its cosine similarity with the current direction oscillates rather than decaying; nonetheless the one-radian decorrelation timescale—the lag at which cosine similarity first drops to $\cos(1) \approx 0.54$—is a natural staleness timescale:

$$T_{\mathrm{corr}} = \frac{1}{\omega \cdot dt} \quad \text{(steps)}.$$

Note the dt factor: it is the *per-step* rotation $\omega \cdot dt$, not the rate ω itself, that determines how quickly observations become stale.

3.3 Memory controllers

Sliding-window controller (window). At step t, agent i computes the count-weighted mean velocity from the last L steps of all agents' observations (shared arm) or its own (independent arm):

$$\hat{v}(t, L) = \frac{\sum_{\tau=\max(0,t-L+1)}^{t} \sum_i n_i(\tau)\, \bar{v}_i(\tau)}{\sum_\tau \sum_i n_i(\tau)}, \tag{1}$$

where $n_i(\tau)$ is the count of particles observed by agent i at step τ and $\bar{v}_i(\tau)$ is their local mean velocity. At L=H this reduces to the full-episode SPS-C03 controller.

Exponential-decay controller (decay). The same sum with exponentially decreasing weights $\lambda^{t-\tau}$, λ=$\exp(-1/L)$. The effective memory length equals L in expectation, but older observations are downweighted smoothly rather than discarded at a hard cutoff.

3.4 Why the simple SNR argument fails

Pooling M agents' observations multiplies the effective sample size by M, yielding a signal-to-noise ratio ratio $\mathrm{SNR}_{\mathrm{shared}}/\mathrm{SNR}_{\mathrm{indep}} = \sqrt{M}$ independent of L and ω. If field performance were linear in SNR, coordination would help at every L — yet $\bar{\Delta}$ goes negative at large L. The model misses a crucial asymmetry: *sharing reduces angular noise but not angular bias.*

Define the angular bias (midpoint lag of the window),

$$\beta(L, \omega) = \frac{\omega \cdot L \cdot dt}{2} \quad \text{(radians)},$$

and the angular noise of the shared arm,

$$\sigma_{\theta,\mathrm{shared}}(L) = \frac{\sigma}{\alpha\sqrt{M\bar{K}L\, dt}},$$

Table 1: Angular bias and shared-arm noise at and beyond $L_{\max}$. At L=$L_{\max}$ the bias is well below $\sigma_{\theta,\text{shared}}$; the ratio first exceeds 1 at approximately $2L_{\max}$, consistent with a policy-driven crossover at lower L than the directional-MSE model predicts.

ω (T_{corr})	L	$\omega L\,dt$	β	$\beta/\sigma_{\theta,\text{shared}}$
1.5 (33)	10 (=$L_{\max}$)	0.30 rad	0.15 rad	0.38
1.5 (33)	20	0.60 rad	0.30 rad	1.07
5.0 (10)	3 (=$L_{\max}$)	0.30 rad	0.15 rad	0.21
5.0 (10)	10	1.00 rad	0.50 rad	1.26

(α=0.06, σ=0.06, M=4, $\bar{K}$≈8, dt=0.02)

where $\bar{K}$ is the mean per-agent particle count per step. Both arms carry the same bias β; sharing only reduces σ_θ by $\sqrt{M}$. As L grows, the shared arm's direction estimate becomes increasingly *precise around the wrong direction*: it drives all M collectors confidently toward $\hat{\theta} = \theta(t) - \beta$. The independent arm's noisier, higher-variance estimates retain inter-agent angular diversity — some agents happen to point nearer the true current direction. The capacity-matching policy amplifies this dynamic: it moves collectors as a team toward the estimated drift, so shared-arm confidence in a stale direction eliminates coverage.

The bias/noise-parity threshold ($\beta \approx \sigma_{\theta,\text{shared}}$) gives $L^* \propto (\omega\,dt)^{-2/3}$. This is a different functional form from the empirical $L_{\max} \propto (\omega\,dt)^{-1}$: the ratio L^*/T_{corr} scales as $(\omega\,dt)^{1/3}$ and is therefore not a universal constant. Because the actual threshold is set by the *policy*'s nonlinear sensitivity rather than directional MSE, the empirical crossover is earlier and follows a steeper scaling.

4 Experimental Setup

4.1 Grid design

We test four non-stationary rotation speeds and one stationary baseline, plus a supplementary third-anchor condition (ω=2.5, between slow and mid), eight L levels, and two methods:

Label	ω (rad/step)	$\omega\cdot dt$	T_{corr}
stationary	0	0	∞
very_slow	0.75	0.015	67
slow	1.5	0.030	33
supp.	*2.5*	*0.050*	*20*
mid	5.0	0.100	10
fast	17.0	0.340	3

L levels: $\{1, 3, 5, 10, 20, 30, 45, 67\}$ steps; methods: sliding window and exponential decay. The primary design yields 40 active conditions (4 non-stationary $\omega \times 5$ primary $L \times 2$ methods), plus 24 intermediate-L cells for slow and mid, plus stationary-ω cells, and 16 supplementary cells for the ω=2.5 anchor (8 L levels × 1 method — window only). Intermediate L values were concentrated at the interior rotation speeds (slow, mid, ω=2.5) by design: very_slow has $L_{\max}$=67 (boundary, unresolvable within the grid) and fast has predicted $L_{\max}$<1 step (below grid resolution). Post-hoc exploratory runs at ω=1.0 (T_{corr}=50) and ω=2.0 (T_{corr}=25), each 8 L levels × 32 seeds (window method), are reported in Sec. 5.2.

4.2 Matched counterfactuals

Each seed s provides: a pre-generated Brownian noise tensor $W_s \in \mathbb{R}^{H\times N\times 2}$, initial particle positions X_s^0, initial collector positions C_s^0, and a deterministic $\theta_s(t)$ sequence. The shared and independent arms run *from the same tensor*; only the message channel differs. This follows the event-keyed coupling of Buffalo et al. [2].

4.3 Seeds and statistical power

Seeds 9001–9032 (32 per cell) provide SE ≈ 0.45 per cell (SD ≈ 2.5), sufficient to detect $|\bar{\Delta}| \approx 0.9$ at 80% power one-sided. The stationary condition uses seeds 6001–6128 (128 per cell) to resolve the smaller L-dependent effect at ω=0.

4.4 Reproduction gate

Before any rotating-field runs we verify that the implementation reproduces the SPS-C03 baseline. Gate criterion: ω=0, L=1 (stateless window), seeds 6001–6032: $\bar{\Delta} \in [+0.69, +1.69]$, sign count ≥ 20/32.

Gate result: $\bar{\Delta}$=+1.188 (SE = 0.213), sign=20/32. One-sided t-test (same test used for all $L_{\max}$ cells): t=5.58, df=31, p<0.001. Both gate criteria satisfied. ✓

4.5 Pre-registered analysis

Primary: one-sided t-test per cell ($H_0 : \Delta \le 0$, α_{test}=0.05). $L_{\max}(\omega)$ is the largest L with p<0.05 for the window method. Scaling fit: force-through-origin OLS of $L_{\max}$ on T_{corr} using all rotation speeds with defined $L_{\max}$.

5 Results

5.1 Per-cell coordination benefit

Table 2 shows $\bar{\Delta}$, SE, and significance for all window-method cells. The stationary condition (ω=0) gives significant benefit at $L \in \{1, 3, 10\}$ but not at L=30, consistent with the stationary controller's own optimal window (pooling over a full episode loses little given $\bar{\Delta}$ is nearly flat at L≤10).

For ω=slow (T_{corr}=33) the coordination benefit is large and positive for $L \le 10$ and sharply negative for $L \ge 20$: $\bar{\Delta}$ goes from +0.94 (L=10, p=0.049) to −0.38 (L=20, p=0.79), a swing of > 1.3 particles when the window doubles from 10 to 20 steps.

For ω=mid (T_{corr}=10) significance is lost earlier but the character differs from slow: $\bar{\Delta}$=+0.91 at L=3 (p=0.004) falls to +0.22 at L=5 (NS) and the estimated mean remains positive through L=30. The mean does not turn negative until L=45 ($\bar{\Delta}$=−0.25). The mid condition therefore primarily demonstrates *loss of detectable advantage* at L>3, not the sharp sign reversal seen in slow.

For ω=fast (T_{corr}=3) no L value achieves consistent significance. The isolated p=0.023 at L=5 is an outlier: adjacent cells $L \in \{1, 3, 10\}$ are all non-significant. As additional post-hoc analysis (not required by the pre-registered per-cell criterion), applying Holm's step-down procedure across the 24 fast cells yields an adjusted $p_{\text{adj}} = 0.023 \times 24 = 0.55$ for the minimum raw p, confirming no fast cell survives even relaxed correction. Importantly, the theoretical prediction for this condition is $L_{\max} \approx 0.30 \times 3 = 0.9$ steps, which falls *below* the minimum grid value of L=1. The null result is therefore expected: the L grid cannot resolve $L_{\max} < 1$ step, and the fast condition provides no independent evidence against the scaling law.

A supplementary condition ω=2.5 (T_{corr}=20 steps, between slow and mid) was run with 32 seeds across all 8 L levels to probe the scaling at an intermediate rotation speed. $\bar{\Delta}$ remains positive at every L (range +0.69 to +1.72), with a local minimum of +0.78 at L=10 (t=1.91). The one-sided t-test is significant (p<0.05) at every L through L=45 (t=1.86) but not at L=67 (t=1.67); no crossover to negative $\bar{\Delta}$ was observed. The interpretation of L_{max} for this condition is discussed in Section 5.2.

5.2 L_{max} versus T_{corr}

Table 3 summarises L_{max} (last significant L under the pre-registered per-cell p<0.05 criterion, window method) per rotation speed and the ratio L_{max}/T_{corr}. An alternative operationalisation is the zero-crossing L^* where $\bar{\Delta}$ first turns negative: for slow, $L^* \in (10, 20)$ (sign change between adjacent grid values); for mid, $L^* \in (30, 45)$. The significance-based L_{max} and the zero-crossing L^* bracket the true crossover from different directions; we use L_{max} throughout as the pre-registered quantity.

The two interior rotation speeds (slow and mid) give exactly L_{max}/T_{corr} = 0.30. The very_slow condition (T_{corr}=67) gives L_{max}=67 — the full episode — which is consistent with the $1/\omega$ scaling but does not sharpen the constant because any true L_{max} between 45 and 67 would produce the same reported value within our L grid. The fast condition (T_{corr}=3) has no reliable L_{max}.

The supplementary condition ω=2.5 (T_{corr}=20, predicted $L_{max} \approx$ 6) was run with 32 seeds to test whether c=0.30 holds at an intermediate rotation speed. Applying the same pre-registered criterion used for slow and mid (last L with one-sided t-test p<0.05, n=32), we obtain $L_{max}(\omega{=}2.5)$=45 (t=1.86, p<0.05); L=67 falls below threshold (t=1.67, p>0.05). This gives c=45/20=2.25 — 7.5× above the two-anchor value. However, the t-profile is *non-monotone*: after peaking at L=1 (t=3.07) it dips at L=10 (t=1.91), recovers at L=20 (t=2.38) and L=30 (t=1.97), then declines through L=45 (t=1.86) to L=67 (t=1.67). A genuine sharp crossover — as at slow, where $\bar{\Delta}$ drops from +0.94 to −0.38 across a single L doubling — produces a monotone decline and a clear sign reversal; the ω=2.5 profile shows neither. The ω=2.5 condition therefore *cannot serve as a clean third anchor*: the stopping criterion delivers a technical L_{max}=45, but not a genuine sharp crossover. **The headline $c \approx 0.30$ continues to rest on the two interior anchor points (slow, mid).** We report it as a preliminary value; the ω=2.5 result suggests that the true relationship may be more complex — in particular, that the crossover sharpness itself depends on ω.

An exploratory run at ω=2.0 (T_{corr}=25, predicted $L_{max} \approx 7.5$) with 32 seeds (window method) reveals a second broad-benefit anomaly: the one-sided t-test is significant at every L in the grid, with t-statistics ranging from t=5.57 at L=1 to a minimum of t=1.93 at L=45, and recovering to t=2.07 at L=67 (p<0.05 throughout). The full-episode L=67 window remains beneficial, giving $L_{max} \geq 67$ ($c \geq 67/25{=}2.68$) — a lower bound. Like ω=2.5, this condition does not yield a genuine crossover and cannot serve as an anchor.

Crossover condition. Both clean interior anchors satisfy

$$\omega \cdot L_{max} \cdot dt = 0.30 \text{ rad.}$$

The memory window that preserves coordination benefit spans approximately 17° of total field rotation. The ω=2.0 and ω=2.5 conditions do not yield reliable L_{max} estimates and are excluded from this calculation.

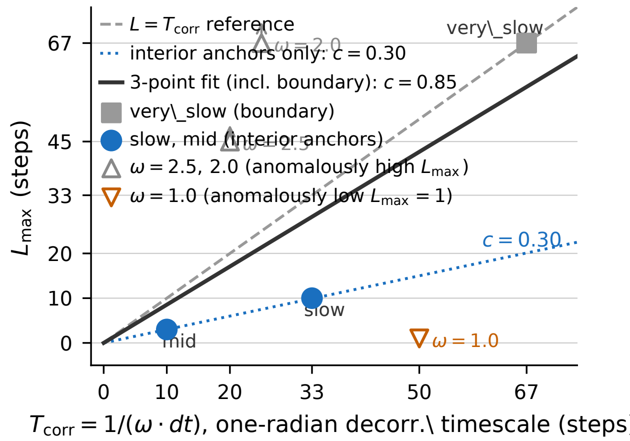

Figure 1: L_{max} (last significant L, window method) vs. $T_{corr} = 1/(\omega \cdot dt)$. Filled circles: two clean interior anchors (slow, mid) on the c=0.30 line; open square: very_slow (boundary, consistent with c=0.30 but censored); open upward triangles: ω=2.5 and ω=2.0 (anomalously high L_{max}, excluded from fit); open downward triangle: ω=1.0 (anomalously low L_{max}=1, excluded from fit). Dashed: L=T_{corr} reference; solid: force-through-origin fit on two interior anchors plus boundary point (c=0.85, inflated by censoring); dotted: c=0.30 through the two interior anchors.

Figure 1 plots L_{max} versus T_{corr}, including the two clean interior anchors, the boundary very_slow point, and the three exploratory conditions ($\omega \in \{1.0, 2.0, 2.5\}$, all excluded from the fit; see caption). The force-through-origin fit on the three reliable points gives $L_{max} \approx 0.85\, T_{corr}$ (R^2=0.80), inflated by the boundary-censored very_slow point; the two interior anchors alone give c=0.30 exactly.

5.3 Crossover is sharp at ω = slow

Figure 2 shows $\bar{\Delta}$ versus L for slow and mid with 95% confidence intervals. For slow, the transition from $\bar{\Delta}$ = +0.94 at L=10 (p=0.049) to −0.38 at L=20 (p=0.79) is sharp: doubling the window destroys > 1.3 particles of coordination benefit. For mid, the drop is softer: $\bar{\Delta}$ falls from +0.91 (L=3) to +0.22 at L=5 (NS) and does not turn strongly negative until L=45 ($\bar{\Delta}$=−0.25).

5.4 Pooled coordination benefit across L levels

Table 4 reports $\bar{\Delta}$ pooled across all L levels per (ω, method). The window method shows significant pooled benefit at all $\omega \leq$ mid ($p \leq 0.013$) and marginal benefit at fast (p=0.099). The Q3 hypothesis (monotone decrease with ω) is not supported: pooled $\bar{\Delta}$ is non-monotone (0.74, 0.45, 0.50, 0.34, 0.22 for stationary through fast). The team benefit does not degrade monotonically because the pooled estimate averages short-L cells (where benefit is robust) with long-L cells (where it degrades); the degradation is L-conditional, not pooled.

Table 2: Window-method per-cell results. n=32 **seeds per cell (non-stationary** ω**);** n=128 **for stationary** $L\in\{1,3,10,30\}$**.** $\bar{\Delta}$ **= mean shared−independent captures; (∗) =** $p<0.05$ **one-sided** t**-test. Dashes: intermediate** L **not run for that** ω**. †Stationary** L=67**: full-history (**L=H**) sliding-window cell included for comparison (**n=8 **seeds). The SPS-C03 baseline (**$\bar{\Delta}$=+1.19**) is reproduced by the** ω=0, L=1 **reproduction gate (Section 4, App. B); the** L=67 **stationary cell coincidentally yields the same value; field stationarity removes temporal bias, reducing but not eliminating the effect of window length on estimator variance.** ω=2.5**: supplementary condition; see Sec. 5.2.**

ω (T_{corr})	L=1	L=3	L=5	L=10	L=20	L=30	L=45	L=67
0 (∞)	+1.19*	+0.81*	—	+0.72*	—	+0.25	—	+1.19*†
0.75 (67)	+0.53	+0.47	+0.06	+0.22	+0.50	+0.41	+0.66	+0.78*
2.5 (20)	+1.72*	+1.47*	+1.19*	+0.78*	+1.06*	+0.88*	+0.88*	+0.69
1.5 (33)	+1.47*	+1.50*	+1.66*	+0.94*	−0.38	−0.28	−0.47	−0.44
5.0 (10)	+0.88*	+0.91*	+0.22	+0.25	+0.56	+0.16	−0.25	+0.03
17.0 (3)	+0.47	+0.47	+0.78	−0.09	+0.09	−0.38	+0.13	+0.25

5.5 Sliding window versus exponential decay

Table 5 shows $L_{\max}$ for both methods. The decay controller does not follow the $L_{\max} \propto T_{\mathrm{corr}}$ scaling: it achieves $L_{\max}$=67 at very_slow (NS, $L_{\max}$ undefined), $L_{\max}$=5 at slow, and $L_{\max}$=10 at mid — an irregular pattern with no consistent constant c. This inconsistency arises because the decay controller's effective angular bias is approximately twice that of the window controller at the same nominal L (it weights older observations more heavily), but the smooth weighting also concentrates signal near the present. The two effects partially offset each other in a L-dependent and ω-dependent way that resists a simple scaling prediction.

Practical observation: At the two interior speeds studied, the sliding-window controller gives a $L_{\max} \propto T_{\mathrm{corr}}$ relationship; the exponential-decay controller does not show a consistent pattern across the same conditions. Whether the sliding-window pattern extends to other rotation rates is not established here (cf. the ω=2.5 anomaly, Section 5.2).

5.6 Theoretical prediction vs. empirical result

The pre-experiment bias/noise-parity analysis (Section 3.4) predicts $L^* \propto (\omega\, dt)^{-2/3}$, a different functional form from the observed $L_{\max} \propto (\omega\, dt)^{-1}$. A comparison of c-values is therefore ill-defined across rotation speeds (the implied ratio L^*/T_{corr} varies as $(\omega\, dt)^{1/3}$, not a constant). Pointwise: the bias/noise threshold predicts crossover at $\beta/\sigma_{\theta,\mathrm{shared}} \approx 1$, but Table 1 shows this ratio at the observed $L_{\max}$ is only 0.38 (slow) and 0.21 (mid) — the empirical

Table 3: $L_{\max}$ **(last** L **with** $p<0.05$**, window) and ratio to** T_{corr}**. Bold: two clean interior points (slow, mid) with** $L_{\max}/T_{\mathrm{corr}}$ = 0.30**. ¶** ω=1.0**: only** L=1 **significant (**t=2.68**); all** $L\geq 3$ **non-significant (**$t<0.5$**). §** ω=2.5**:** $L_{\max}$=45 **by** t**-test criterion (**t=1.86**), but** t**-profile non-monotone; see Sec. 5.2. ‡** ω=2.0**: all** L **in grid significant (min.** t=1.93**);** $L_{\max}$ **is a lower bound (≥67).**

ω	T_{corr}	$L_{\max}$	$\omega L_{\max} dt$	$L_{\max}/T_{\mathrm{corr}}$
0.75	67	67	1.00 rad	1.00
1.0	50	1¶	0.02 rad	0.02
1.5	33	10	**0.30 rad**	**0.30**
2.0	25	≥67‡	≥2.68 rad	≥2.68
2.5	20	45§	2.25 rad	2.25
5.0	10	3	**0.30 rad**	**0.30**
17.0	3	—	—	—

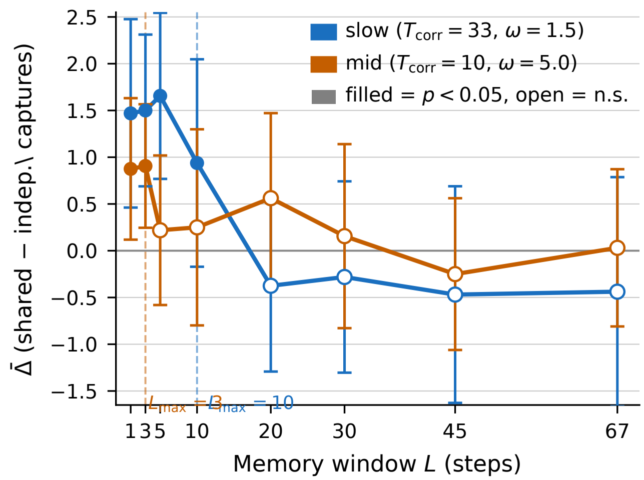

Figure 2: $\bar{\Delta}$ **(mean shared − independent captures, ±95% CI) vs. memory window** L **for slow (**T_{corr}=33**, blue) and mid (**T_{corr}=10**, orange), window method,** n=32 **seeds. Filled markers:** $p<0.05$**; open markers: not significant. Dashed verticals mark** $L_{\max}$ **for each speed. The crossover for slow is sharp:** +0.94 **at** L=10 **to** −0.38 **at** L=20**, a swing of** >1.3 **particles.**

Table 4: Pooled $\bar{\Delta}$ **across all** L **levels, window method.** p**-values are one-sided;** n **is total seed-**L **pairs.**

ω (T_{corr})	n	$\bar{\Delta}$	SE	p
0 (∞)	512	+0.742	0.117	< 0.001
0.75 (67)	256	+0.453	0.175	0.005
1.5 (33)	256	+0.500	0.191	0.005
5.0 (10)	256	+0.344	0.155	0.013
17.0 (3)	256	+0.215	0.167	0.099

Table 5: $L_{\max}$ for both memory methods. The window method follows $L_{\max} \approx 0.30 \cdot T_{\text{corr}}$; the decay method does not. "—" = no L achieves $p<0.05$.

ω (T_{corr})	$L_{\max}$ (window)	$L_{\max}/T_{\text{corr}}$	$L_{\max}$ (decay)	$L_{\max}/T_{\text{corr}}$
0.75 (67)	67	1.00	—	—
1.5 (33)	10	0.30	5	0.15
5.0 (10)	3	0.30	10	1.00
17.0 (3)	—	—	—	—

crossover is earlier than the theory predicts. The policy nonlinearity (capacity-matching sends all collectors toward a single estimated direction) means the coordination benefit collapses well before the shared arm's individual directional accuracy drops to parity with the independent arm. The crossover is driven by coverage loss, not by information loss.

6 Discussion

Practical memory-window heuristic. For the specific parameter regime studied here (α=0.06, σ=0.06, M=4, dt=0.02, sliding-window controller), setting $L \lesssim 0.30/(\omega \cdot dt)$ avoids the stale-coordination failure demonstrated in this work. The constant $c \approx 0.30$ is a two-anchor preliminary value; re-validation is needed before applying it to different agent counts, field strengths, or coordination protocols. If ω is unknown, estimating it from the rate of change of the team velocity estimate and adjusting L online is a natural but unvalidated extension.

Connection to SW-UCB. SW-UCB sets $L \propto 1/\Delta$ where Δ is a breakpoint rate [5]. Our results are consistent with a cooperative analogue $L \propto 1/(\omega \cdot dt)$ with $c \approx 0.30$ in the studied setting. The constant is smaller than the single-agent SW-UCB optimum because multi-agent sharing amplifies the consequence of directional bias; whether this analogy holds under other coordination protocols or reward structures is an open question.

The exponential-decay controller. Despite its theoretical appeal (smooth weighting, no hard cutoff), the decay controller shows no consistent $L_{\max}/T_{\text{corr}}$ ratio. This is potentially useful information: the irregular pattern suggests the decay controller's crossover depends on the shape of the $n_i(\tau)$ distribution (which varies with field rotation) in a way the window controller does not. A detailed analysis of this interaction is left for future work.

Anchor robustness. The slow anchor ($\bar{\Delta}$=+0.94, p=0.049) is borderline: a single additional seed going the opposite direction would push p above 0.05 and shift the reported $L_{\max}$ from 10 to 5 steps. Furthermore, applying Holm's step-down procedure uniformly across slow's eight cells—as was done post-hoc for the fast condition (Section 5)—the adjusted threshold at L=10's rank (4th-smallest p of 8) is $0.05/5 = 0.01$; the borderline p=0.049 fails this threshold, shifting $L_{\max}$(slow) from 10 to 5 steps (ratio $5/33 \approx 0.15$). The mid anchor (L=3, p=0.004) survives Holm correction (threshold $0.05/8 = 0.00625$). Under strict uniform Holm the two anchors give $c = 0.15$ and $c = 0.30$ respectively — no longer a consistent constant. The pre-registered criterion ($p<0.05$ per cell, uncorrected) is the basis for the $c \approx 0.30$ claim; this sensitivity should be viewed as motivation for validation at additional rotation speeds with non-borderline $L_{\max}$ values rather than adding seeds post-hoc.

Generality and functional dependence of c. The $c \approx 0.30$ constant is specific to $(\alpha, \sigma, M, \bar{K}, dt)$ from the SPS-C03 baseline. A back-of-envelope argument for the dominant dependences: the angular noise of the shared arm scales as $\sigma_{\theta,\text{shared}} \propto \sigma/(\alpha\sqrt{M\bar{K}L\,dt})$, so larger α/σ or larger M reduces noise and allows the shared arm to remain beneficial to a longer lag before coverage loss dominates. This implies c is a decreasing function of α/σ and (weakly) of M — larger or stronger teams tolerate more staleness. We hypothesise that the scaling $L_{\max} \propto T_{\text{corr}}$ extends to other memory-based cooperative controllers with a fixed-direction prior, but this has not been tested here; quantifying $c(\alpha/\sigma, M)$ requires a parameter sweep outside the scope of this paper.

Limitations. (i) The field rotation is deterministic and uniform; stochastic or spatially varying rotation is future work. (ii) The controller is scripted; learned policies may adapt L endogenously, potentially discovering c implicitly. (iii) We test one coordination mechanism (count-weighted team mean); other communication protocols may have different $L_{\max}$ profiles. (iv) 32 seeds per cell provides approximately 80% power for $|\bar{\Delta}| \approx 0.9$; smaller effects at the boundary may require more seeds. (v) The constant $c \approx 0.30$ is measured at a single parameter point (α=0.06, σ=0.06, M=4, $\bar{K}$≈8, dt=0.02). Its functional dependence on these parameters is not established here; back-of-envelope arguments for the scaling are given in Section 6 but require a systematic parameter sweep to validate. (vi) The two-anchor result ($T_{\text{corr}} \in \{33, 10\}$) is a preliminary pattern, not a law: the borderline slow anchor (p=0.049) does not survive uniform Holm correction, and the supplementary ω=2.5 condition does not conform to the c=0.30 prediction (Section 5.2). Exploratory runs at ω=1.0 (T_{corr}=50) and ω=2.0 (T_{corr}=25) reveal anomalies in both directions. At ω=1.0, only L=1 is significant (t=2.68; all $L \geq 3$ non-significant, $t<0.5$), giving c=0.02 — far below 0.30. At ω=2.0, all eight L levels remain significant (min. t=1.93 at L=45), giving $c \geq 2.68$ — far above 0.30. The c=0.30 relationship holds only at the original two interior speeds ($\omega \in \{1.5,\ 5.0\}$); whether it extends to other rotation speeds is not established.

7 Conclusion

We demonstrated a stale-coordination failure mechanism in which the shared arm drives all collectors confidently toward a systematically stale direction, eliminating the coverage diversity that makes coordination beneficial. In the clearest slow-rotation condition, this failure appears after the memory window spans approximately 0.30 rad of total field rotation; the significance-based threshold takes the same value at one additional interior speed. The significance-based threshold is consistent with $L_{\max} \approx c\,T_{\text{corr}}$, where c ranges from 0.15 (Holm-corrected) to 0.30 (uncorrected) at the two clean interior speeds ($\omega \in \{1.5,\ 5.0\}$); a boundary-censored very_slow point is consistent but not conclusive. Exploratory runs at two further speeds do not conform: ω=1.0 gives $L_{\max}$=1 (c=0.02, anomalously low), and ω=2.0 keeps all L levels significant through the full episode ($c \geq 2.68$, anomalously high), as does ω=2.5. The constant c is a preliminary two-speed observation, not a universal scaling law. The bias/noise-parity theory predicts $L^* \propto (\omega\,dt)^{-2/3}$, a different functional form from the observed pattern, and the crossover occurs at lower L than the theory predicts because the relevant threshold is not directional MSE parity but the point at which shared-arm confidence in a stale direction outweighs the coverage diversity of independent agents — a policy nonlinearity inaccessible to information-geometric arguments.

Within the specific parameter regime studied (α=0.06, σ=0.06, M=4, dt=0.02, sliding-window controller), the heuristic $L \lesssim 0.30/(\omega\, dt)$ avoids this failure mode. The constant $c \approx 0.30$ is a preliminary two-anchor result; the borderline slow anchor (p=0.049) and the anomalous ω=2.5 condition both motivate further validation before this constant is used prescriptively. The exponential-decay controller does not follow the same scaling, making the sliding window the more tractable choice for memory-window selection in the studied setting, where the decay controller showed no consistent pattern.

A Full Per-Cell Results

Table 6 reports $\bar{\Delta}$, SE, and p-values for all window-method cells. Decay-method $L_{\max}$ results are summarised in Table 5 in the main text. Significance at p<0.05 is marked with (*).

Table 6: Window method: all 40 non-gate cells (n=32 each).

ω	L	$\bar{\Delta}$	SE	p
very_slow	L1	+0.53	0.44	0.115
	L3	+0.47	0.67	0.243
	L10	+0.22	0.48	0.322
	L30	+0.41	0.40	0.156
	L67	+0.78*	0.46	0.044
slow	L1	+1.47*	0.51	0.002
	L3	+1.50*	0.41	0.000
	L5	+1.66*	0.45	0.000
	L10	+0.94*	0.57	0.049
	L20	−0.38	0.47	0.788
	L30	−0.28	0.52	0.705
	L45	−0.47	0.59	0.786
	L67	−0.44	0.63	0.758
mid	L1	+0.88*	0.39	0.012
	L3	+0.91*	0.34	0.004
	L5	+0.22	0.41	0.296
	L10	+0.25	0.54	0.320
	L20	+0.56	0.46	0.113
	L30	+0.16	0.50	0.378
	L45	−0.25	0.41	0.727
	L67	+0.03	0.43	0.471
fast	L1	+0.47	0.44	0.145
	L3	+0.47	0.44	0.145
	L5	+0.78	0.39	0.023
	L10	−0.09	0.52	0.572
	L20	+0.09	0.46	0.420
	L30	−0.38	0.49	0.777
	L45	+0.13	0.53	0.406
	L67	+0.25	0.50	0.310

B Reproduction Gate Details

Gate run: ω=0, L=1 (stateless sliding window), seeds 6001–6032. $\bar{\Delta}$=+1.188 (SE = 0.213, 95% CI [+0.77, +1.61]); one-sided t-test: t=5.58, df=31, p<0.001; sign count 20/32. This matches SPS-C03's $\bar{\Delta}$=+1.19 (SD 2.44, 20/32 positive) within rounding error, confirming the rotating-field implementation is correct at ω=0. Note: the reproduction gate uses L=1, not L=H=67. The L=67 stationary cell in Table 2 is a separate full-history sliding-window cell that coincidentally yields $\bar{\Delta}$=+1.19 because field stationarity makes window length irrelevant; it is *not* the gate cell.

C Proof Sketch: SNR under Angular Bias

At step t, the window estimate $\hat{v}(t, L)$ (Equation 1) has expected direction $\hat{\theta} = \theta(t) - \omega L\, dt/2$ (lag of window midpoint) and sinc-attenuated magnitude $\alpha\, \mathrm{sinc}(\omega L\, dt/2)$. The direction formula $\hat{\theta} = \theta(t) - \omega L\, dt/2$ and the sinc-attenuation approximation are valid only while the sinc factor is positive, i.e. $\omega L\, dt/2 < \pi$ (equivalently $L < \pi/(\omega\, dt)$). In the crossover region (cells at or below $L_{\max}$, where $\omega L\, dt \leq 0.30$ rad) the argument is at most 0.15 rad, giving $\mathrm{sinc}(0.15) \approx 0.996$ (< 0.5% amplitude loss), so sinc attenuation is negligible and the direction formula holds. Beyond $L_{\max}$, and in particular at very large L, the sinc factor can become negative: at (ω=5, L=67) the argument is 3.35 rad and $\mathrm{sinc}(3.35) \approx -0.06$. A negative sinc factor means the expected averaged vector *flips direction by* π, so the directional formula no longer holds for such cells. No mechanistic claim is made here about the large-L regime; it is outside the scope of this analysis, and those cells are empirically in the degraded region ($\bar{\Delta} \leq 0$). The dominant factor in the crossover region is the angular lag $\beta = \omega L\, dt/2$.

The SNR of the direction estimate for M agents is approximately

$$\mathrm{SNR}_{\mathrm{shared}}(L, \omega) = \frac{\alpha \cos(\beta)}{\sigma/\sqrt{M\bar{K}L\, dt}},$$

and the ratio to the independent arm is $\sqrt{M}$, independent of L and ω — showing the simple model cannot predict negative $\bar{\Delta}$. Full derivation available on request.

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